

**ShortStay – Online Marketplace for Short-Term
Accommodations
Interim Report
Academic Year 2021/2022**

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Group 02

CIS3012 - Group Project

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Project Details

Title of Project

ShortStay – Online Marketplace for Short-Term Accommodations

Group Members' Details

Table 1 - Group Member Details

Name	Email Address / Phone Number	Index number	Registration Number	Role of the Project
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Supervisor/Advisor Details

Supervisor's name - Mr. K.D. Tharindu Rangana

Supervisor's email - kdtrangana@sjp.ac.lk

1 Introduction

ShortStay is a web-based platform that connects guests with hosts who provide short-term accommodation. The system also includes administrative, financial, and verification roles to ensure trust and smooth operation.

1.1 Motivation for Project

Sri Lanka is a well-known tourist destination. Many local property owners want to earn extra income by renting their spaces. However, existing platforms like Airbnb are not very user-friendly for local users.

ShortStay is designed to:

- Support local users
- Provide a simple interface
- Ensure trust through verification

1.2 Objectives of the Project

Main Objective

To develop a reliable and user-friendly short-term accommodation platform.

Specific Objectives

- Implement 5 user roles (Guest, Host, Admin, Payment Manager, Field Inspector)
- Enable booking and payment system
- Provide a property verification process
- Ensure a secure and efficient system

1.3 Scope of the Project

The system will:

- Allow users to register and log in
- Allow hosts to list properties
- Allow guests to search and book
- Handle payments and refunds
- Verify properties using inspectors

The system will not include:

- Physical transportation services
- Long-term rentals

1.4 Product Perspective

ShortStay is a standalone web application. It may integrate with:

- Payment gateways
- Email notification systems

1.5 User Classes and Characteristics

Table 2 - User Classes and Characteristics

User Type	Description
Guest	Books accommodations
Host	Lists properties
Admin	Manages system
Payment Manager	Handles payments
Field Inspector	Verifies properties

1.6 Design and Implementation Constraints

- Web browser compatibility required
- Must operate over HTTPS
- Limited time and resources
- Data security and role-based access control enforced

1.7 Assumptions and Dependencies

- Email APIs and payment gateway APIs are available and reliable
- Hosts provide correct information
- Users have basic digital literacy

1.8 Specific Requirements

The ShortStay system includes a set of core features and role-specific functionalities to ensure smooth operation of the platform. The system is designed using a role-based approach to manage different responsibilities effectively.

1.8.1 Core Features

- User registration and authentication with role-based access
- Secure login and logout functionality
- Profile management for all users
- Property listing and management system
- Advanced search and filtering options (location, price, availability, ratings)
- Booking system with date selection and confirmation
- Online payment processing
- Notification system
- Review and rating system
- Property verification mechanism
- Role-based dashboards
- System monitoring and reporting

1.8.2 Guest Functionalities

- Search for properties using filters such as location, price, and dates
- View detailed property information (images, description, price, ratings)
- Make bookings and select dates
- Perform online payments
- Cancel bookings based on policies
- View booking history
- Submit reviews and ratings for completed stays
- Receive email notifications related to bookings

1.8.3 Host Functionalities

- Register and log in to the system
- Create and manage property listings
- Upload property details including images, price, and availability
- Submit properties for verification
- Update or remove property listings
- View booking requests and confirmations
- Manage availability calendar
- View earnings and booking history
- Receive email notifications regarding bookings and verification status

1.8.4 Admin Functionalities

- Manage all users (Guest, Host, Payment Manager, Field Inspector)
- Approve or reject property listings after verification
- Monitor system activities
- Handle complaints and disputes
- Generate reports (user activity, bookings, system performance)
- Suspend or remove users who violate policies
- Ensure overall system functionality and security

1.8.5 Payment Manager Functionalities

- Monitor all payment transactions
- Process payments from guests
- Manage payouts to hosts
- Handle refunds and cancellations
- Track transaction history
- Generate financial reports
- Identify and manage failed or pending transactions

1.8.6 Field Inspector Functionalities

- View list of properties pending verification
- Schedule and conduct property inspections
- Verify property details such as location, facilities, and safety
- Capture and upload inspection reports and images
- Approve or reject properties based on standards
- Submit verification results to Admin
- Maintain inspection history records

1.9 Functional Requirements

- Guests can search and book properties
- Hosts can manage listings
- Admin can manage the system
- Payment Manager handles transactions
- Field Inspector verifies properties

1.10 Non-Functional Requirements

- User-friendly interface with intuitive navigation
- System supports a minimum of 100 concurrent users
- Uptime of at least 99.5%
- Operate across all major browsers
- Maintain clean and maintainable code for future updates
- Comply with relevant data protection laws (e.g., GDPR)

1.11 Deliverables

- A fully functional web-based application
- Database with all required data structures
- Complete project source code
- Payment integration module
- Technical documentation
- Demo or presentation of the system
- User training or a help guide to operate the system

1.12 Resource Requirements

1.12.1 Software Requirements

- Frontend: React
- Backend: Node.js / JavaScript
- Database: MySQL
- Payment Gateway (PayHere / Stripe)
- Email API for notifications
- Web browser (Google Chrome)
- Microsoft Office for documentation

1.12.2 Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: 4GB minimum (8GB recommended)
- Storage: 256GB SSD or higher
- Stable Internet connection

1.13 Proposed Method for Evaluating Success

- Unit Testing: Each module will be tested individually
- Integration Testing: Modules tested together
- System Testing: Full system validation
- User Feedback: Feedback from sample users
- Supervisor Evaluation: Final academic evaluation

1.14 Proposed Client/Beneficiary

This project is designed for academic purposes at the University of Sri Jayewardenepura. However, it can mainly benefit local property owners, tourists, and travelers. It also supports the Sri Lankan tourism industry by providing an easy-to-use platform for short-term accommodation and additional income opportunities.

1.15 Chapter Summary

This chapter provided an overview of the ShortStay project, including its objectives, scope, user roles, and requirements. It helps to clearly understand the system before moving to the design and development stages. The chapter provides a strong foundation for the methodology, analysis, and system design discussed in subsequent chapters.

2 Methodology

2.1 Introduction

Methodology defines the approach, techniques, tools, and processes followed to build a reliable and maintainable system. For the development of this short-term accommodation marketplace, methodology ensures that all aspects, from requirement gathering to deployment, are structured, controlled, and measurable.

An accommodation marketplace involves multiple user roles (guests, hosts, admins), complex workflows like booking management, pricing, and reviews, and third-party integrations such as payment gateways and map services. A well-defined methodology ensures that the system,

- Provides a seamless experience for both guests and hosts
- Integrates core functionalities like listing management, booking, payments, and reviews
- Follows industry-standard development practices
- Ensures security, reliability, and maintainability

2.2 Data Collection Methods

2.2.1 Observations

The team observed existing short-term rental platforms (Airbnb, Booking.com) to study their operational workflows. Key observations are :

- **Guest booking flow:** How guests search, filter, and book accommodations
- **Host listing management:** How hosts create listings, set pricing, and manage availability calendars
- **Review system:** How guests and hosts leave and respond to reviews post-stay
- **Admin moderation:** How platforms handle disputes, flagged listings, and user verification

Outcome: Observations revealed user pain points such as unclear availability calendars, slow booking confirmations, and insufficient trust mechanisms, forming the basis for feature prioritization.

2.2.2 Secondary Research

The team reviewed existing platforms, academic literature, and industry standards to gather insights into:

- Common marketplace architectures (multi-tenant, microservices)
- Technologies commonly used (React, Node.js, PostgreSQL/MySQL)
- Integration of map APIs (Google Maps) and payment gateways
- Legal and compliance considerations for short-term rental platforms (GDPR, host regulations)

2.3 Software Process Model

The Agile Software Development Model was selected due to the dynamic and iterative nature of marketplace development, where user feedback and feature changes are frequent.

2.3.1 Justification for Agile

- Supports incremental delivery of features (search, booking, payments) in separate sprints
- Allows rapid adaptation when requirements evolve based on user testing
- Enables continuous integration and testing across frontend and backend modules
- Facilitates regular feedback from the supervisor and sample users

2.3.2 Agile Development Phases Followed

- Requirement Analysis - Gathered through observations, questionnaires, and competitor analysis; defined functional/non-functional requirements for all roles
- System Design - Developed ER diagrams, database schema, UI mockups in Figma; designed booking flows, listing management, and payment processes
- Implementation - Frontend built with React; backend APIs with Node.js; relational database in MySQL
- Testing - Unit, integration, and system testing conducted each sprint
- Deployment & Feedback - System prepared for demo; feedback collected for improvements

2.4 Software Development Tools

2.4.1 Frontend Development Tools

- **React.js** – Framework for building dynamic and responsive user interfaces
- **HTML, CSS, JavaScript** – Core technologies for web interface development

2.4.2 Backend Development Tools

- **Node.js / Express.js** — RESTful API development for business logic
- **JWT** — Secure authentication and role-based access control

2.4.3 Database Tools

- **MySQL** – Relational database management system
- **MySQL Workbench** – Tools for database management, creating tables, relationships, and executing queries

2.4.4 Other Tools

- Payment Gateway (PayHere / Stripe) - Secure guest payments and host payouts
- Google Maps API - Location-based listing search and map display
- Email API (SendGrid / NodeMailer) - Booking confirmations, reminders, and host notifications
- GitHub / Git - Version control and team collaboration
- Visual Studio Code - Primary IDE

2.5 Testing Strategies

- Unit Testing - Individual modules (search, booking, payment) tested independently
- Integration Testing - Verified seamless interaction between frontend, backend, and database
- System Testing - End-to-end testing across all user roles and booking scenarios
- User Acceptance Testing (UAT) - Sample guests and hosts evaluated usability and functionality
- Performance Testing - Ensured the system handles concurrent searches and bookings efficiently

Testing Tools: Postman (API testing), Browser Developer Tools (frontend), Manual Testing (usability)

2.6 Implementation Plan

Table 3 - Implementation Plan

Phase	Activities
Phase 1: Frontend Development	Completed dashboards and pages for Guest, Host, and Admin roles, applied responsive design
Phase 2: Database Design	Created tables for users, listings, bookings, payments, and reviews
Phase 3: Backend Development	Implemented APIs, authentication, search/filter logic, and booking management
Phase 4: Integration	Connected frontend with backend and database; tested CRUD operations and booking flow.
Phase 5: Testing & Optimization	Conducted unit, integration, system, and UAT testing; resolved bugs
Phase 6: Final Deployment & Documentation	Prepared for demonstration; created user manuals and technical documentation

2.7 Chapter Summary

This chapter outlined the methodology adopted for ShortStay (short-term accommodation marketplace), covering data collection via observations and questionnaires, adoption of the Agile model, selection of development tools, testing strategies, and a phased implementation plan. By following this approach, the team ensures a user-friendly, secure, and scalable platform meeting both guest and host requirements.

3 Analysis

3.1 Introduction

The system analysis phase is crucial for converting collected requirements into a coherent and structured system design. It centers on grasping how the suggested system operates, the arrangement of its components, and the ways various users engage within the system. This chapter provides an analysis of ShortStay (Online Marketplace for Short-Term Accommodations), utilizing recognized modeling techniques like Unified Modeling Language (UML) and Entity-Relationship (ER) diagrams.

The ShortStay platform is created to accommodate various user roles, such as Guests, Hosts, Admins, Payment Managers, and Field Inspectors. Every one of these roles engages with the system uniquely, making it crucial to explicitly outline their actions and duties during this stage. The purpose of this analysis is to,

- Offer an organized perspective on system features and operations.
- Depict user engagements and processes within the system.
- Recognize system elements and their connections
- Create a dependable and effective database framework.
- Make certain that all outlined requirements are adequately addressed before starting development

To accomplish this, different diagrams like Use Case Diagrams, Class Diagrams, and ER Diagrams are employed. These diagrams together provide a thorough depiction of the system, laying a solid groundwork for the implementation stage and aiding in reducing mistakes during development.

3.2 Use Case Diagram of Proposed System

The Use Case Diagram provides a high-level view of system functionality and ensures that all user interactions are clearly defined.

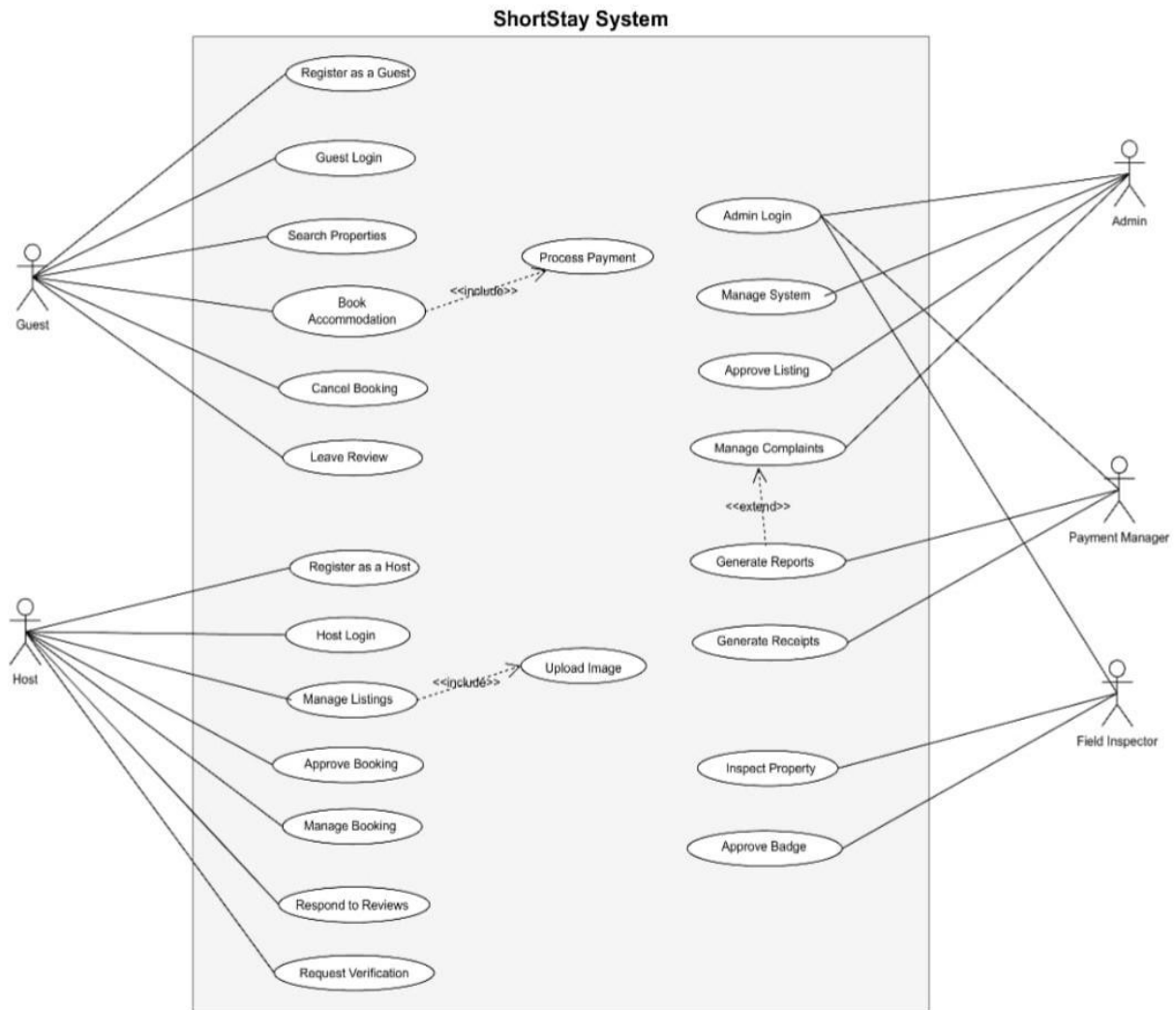


Figure 1 - Use Case Diagram

3.3 Class Diagram of Proposed System

The Class Diagram represents the static structure of the ShortStay system by defining its classes, attributes, methods, and relationships.

ShortStay System - UML Class Diagram

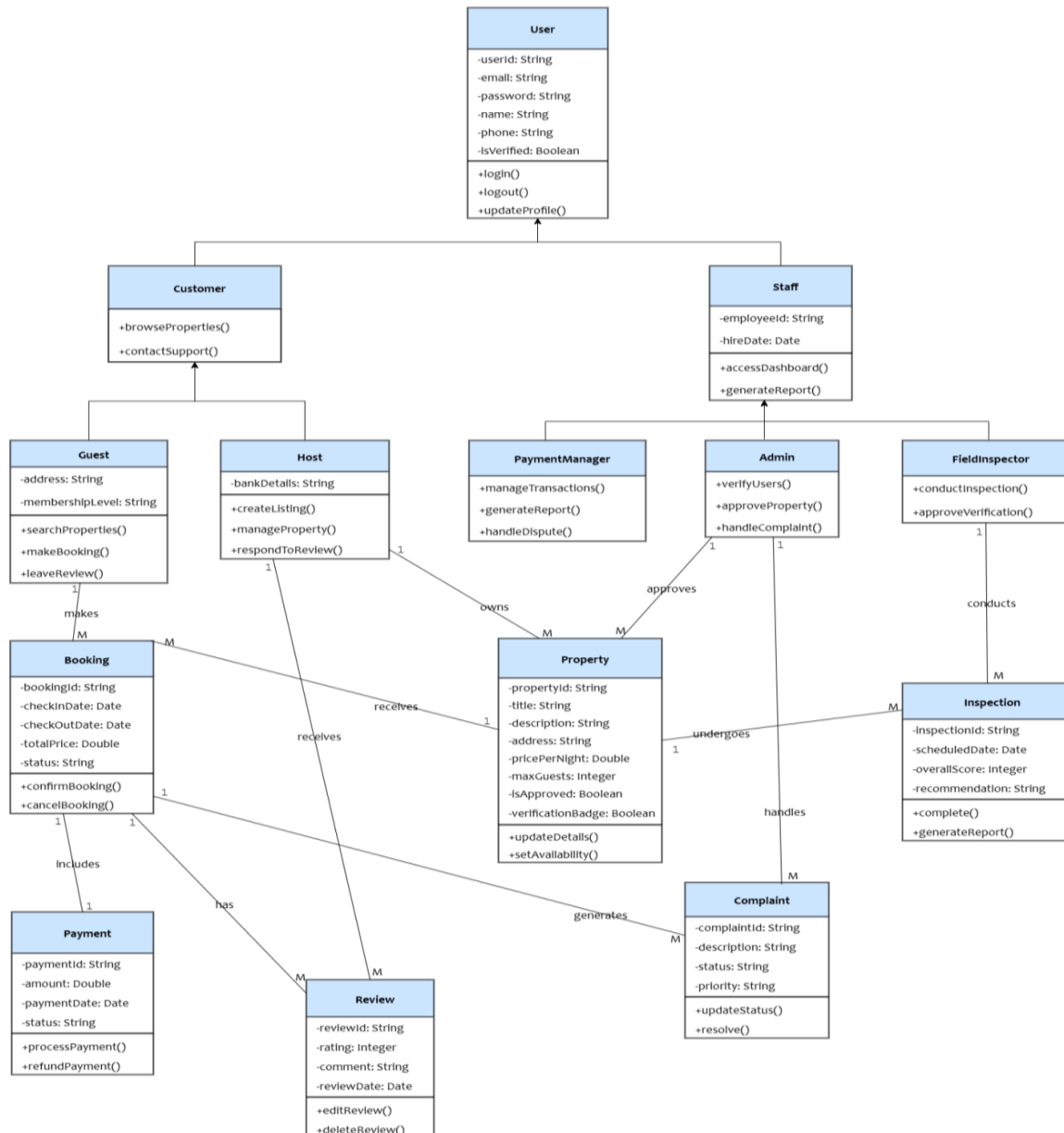


Figure 2 - Class Diagram

3.4 ER Diagram of the Proposed System

The Entity-Relationship (ER) Diagram represents the database structure of the ShortStay system by defining entities, attributes, and relationships.

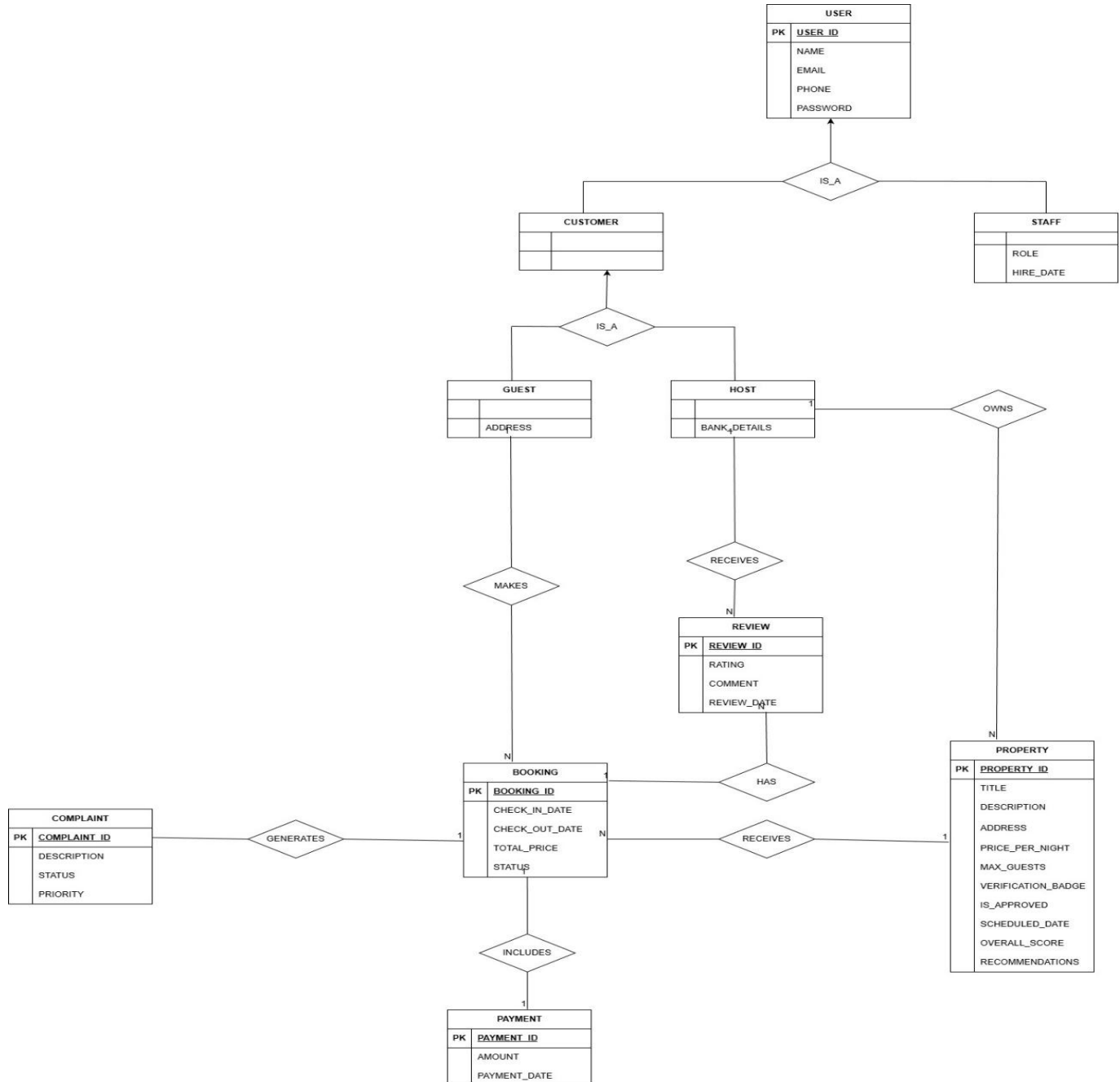


Figure 3 - ER Diagram

3.5 Chapter Summary

This chapter provided a detailed analysis of the ShortStay system using various modeling techniques.

- The Use Case Diagram illustrated the interactions between users and the system.
- The Class Diagram defined the system's structure.
- The ER Diagram described the database design.

Together, these diagrams provide a comprehensive blueprint for the development phase, ensuring that all system requirements are accurately represented and implemented. The analysis phase plays a vital role in minimizing errors and improving the overall quality of the system.

4 Solution Design

4.1 Introduction

The process of designing solutions involves turning the system's requirements and models into a real system architecture. The process of designing the frontend solutions for the ShortStay Smart Rental Platform includes:

- Creating interfaces for each role (Guest, Host, Admin)
- Completing UI/UX design in Figma
- Ensuring consistency and usability
- Preparing the system for backend integration

This chapter will present an overall look at the solution design and its components, such as the frontend architecture, UI design, navigation, and system modules.

4.2 Role-Based Dashboard Design

The ShortStay system includes five dashboards, each tailored to specific user roles.

Table 4 - Role-Based Dashboard Design

Role	Key Features
Guest	Search properties, book stays, make payments, leave reviews
Host	Manage listings, update availability, handle bookings, communicate with guests
Admin	Manage users, monitor system, handle complaints, view analytics
Payment Manager	Track transactions, manage payouts, generate financial reports
Field Inspector	Approve listings, verify users, detect fraudulent content

4.3 UI Design with Figma

The UI for all the frontend screens has been designed using Figma to achieve consistency and usability.

4.3.1 Key Design Principles Applied:

- Consistency: Consistent color schemes, fonts, icons, and button design on all screens
- Accessibility: Readable text with high contrast ratios and friendly input fields

- **Responsive Design:** Flexible layouts that accommodate different screen resolutions through media queries
- **Feedback from User:** Real-time feedback using alerts, bookings confirmation, and errors

4.3.2 Screens Included in Figma Design:

There are certain screens present in the Figma design of ShortStay, which are listed below:

- Homepage (search bar, featured listings)
- Login and Registration Screens
- Search Result Screen (filtering and sorting options)
- Property Detail Screen (images, features, reviews, booking form)
- Guest Dashboard (my reservations)
- Host Dashboard (listings, reservations, earnings)

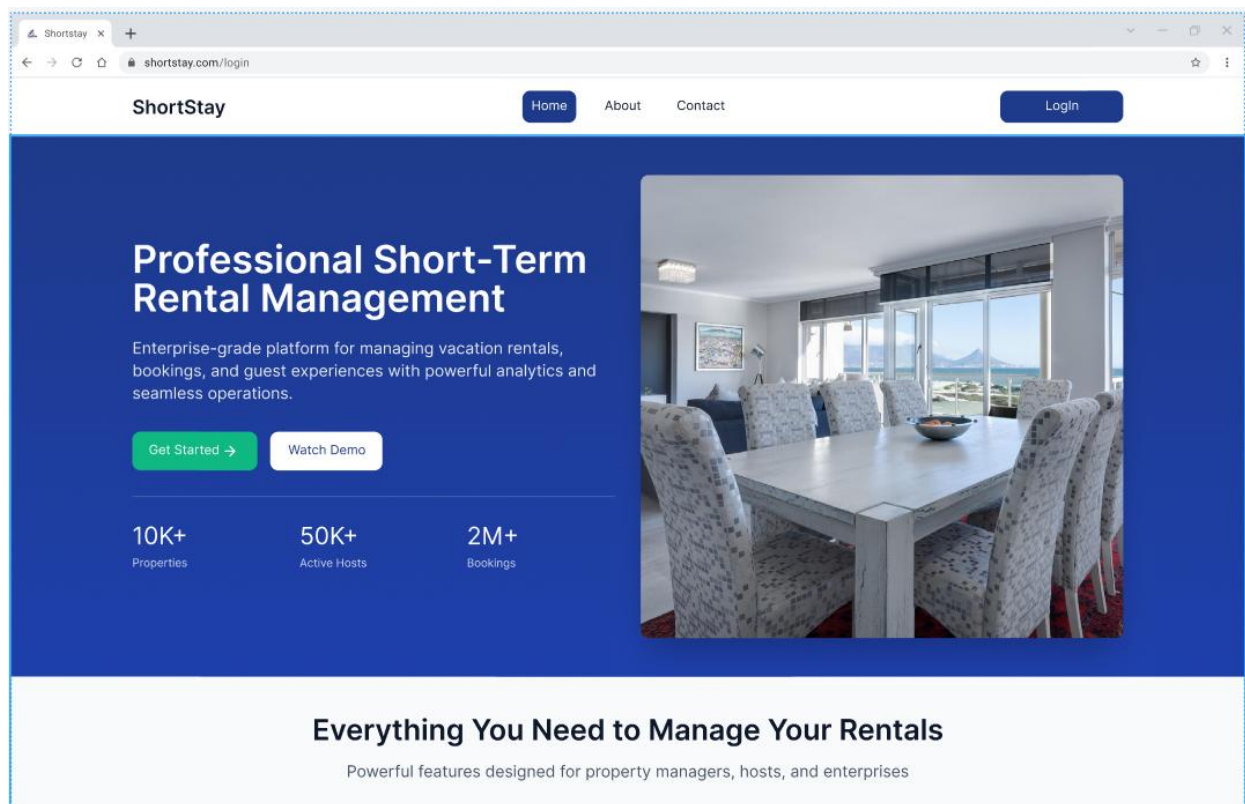


Figure 4 - Home Page

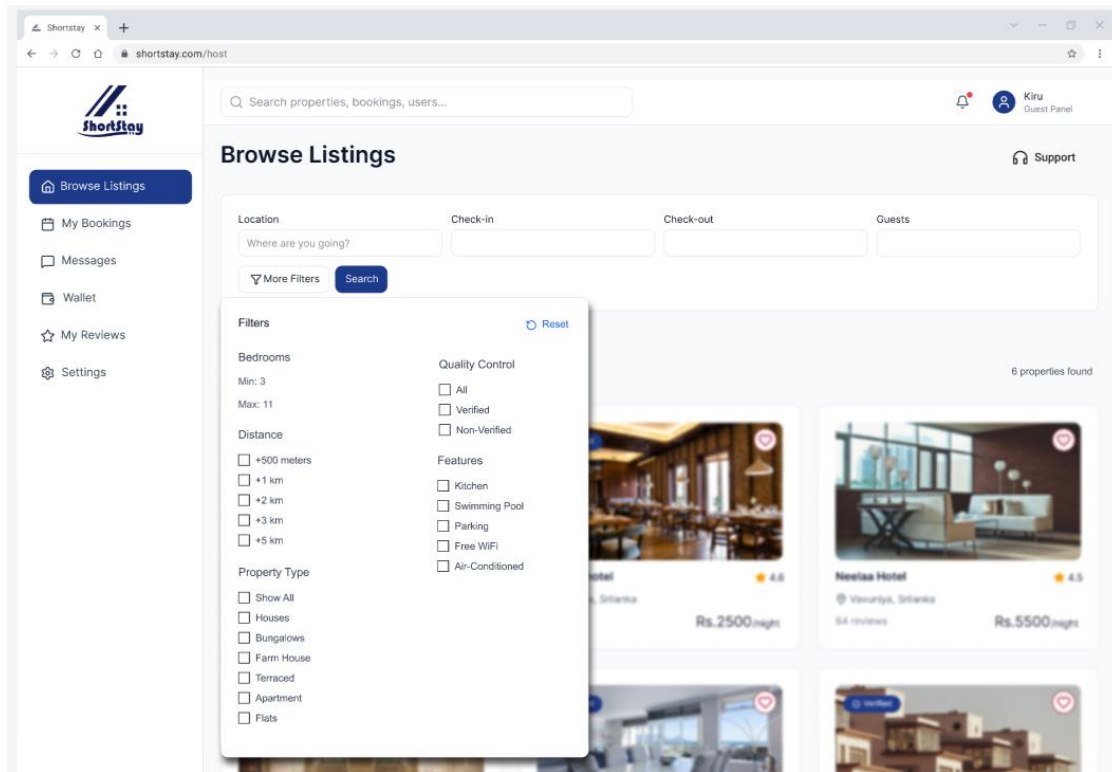


Figure 5 - Searching and Filtering

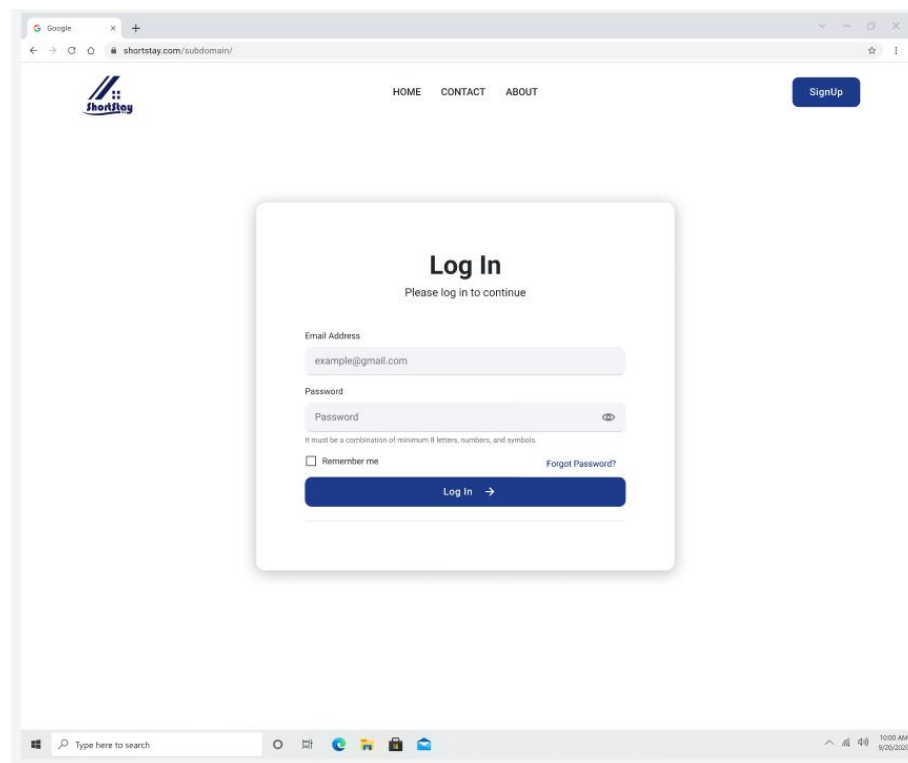


Figure 6 - Log In Page

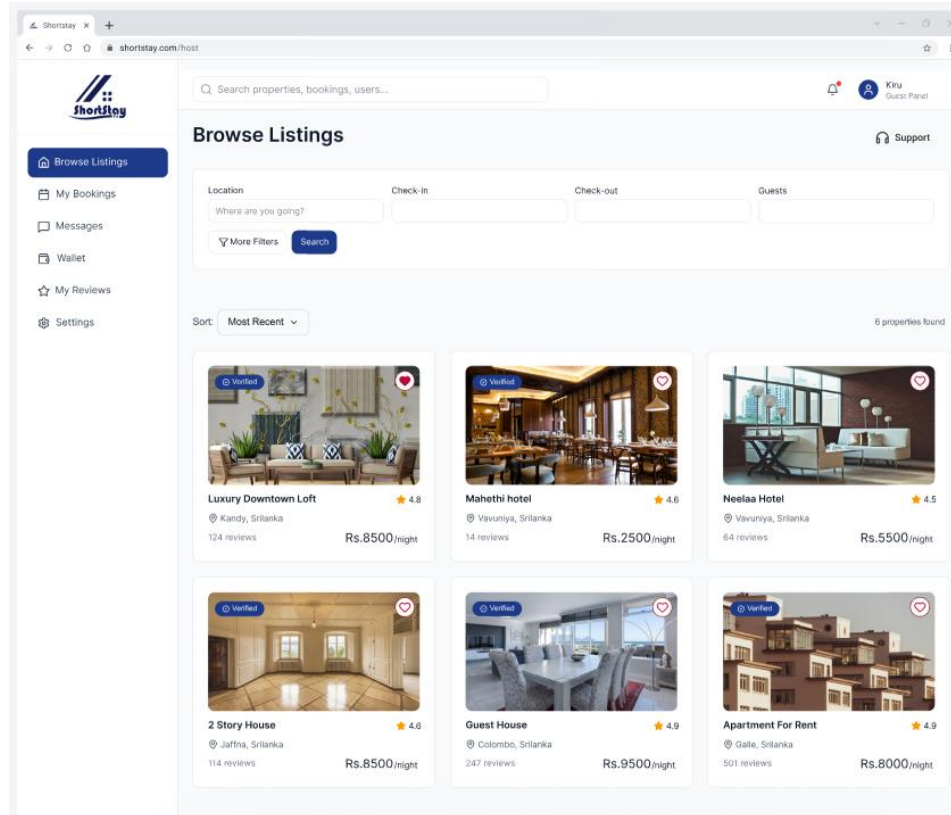


Figure 7 - Guest Dashboard

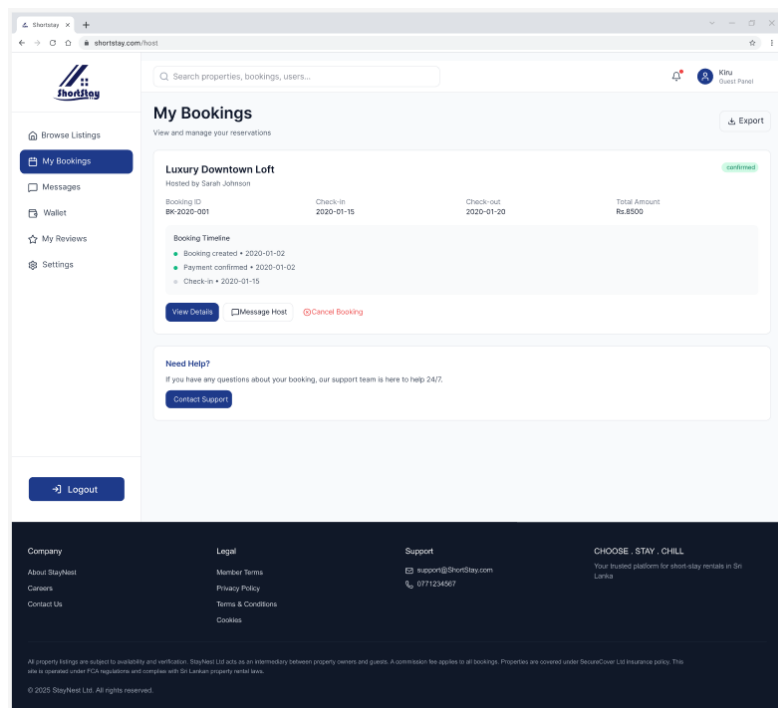


Figure 8 - My bookings

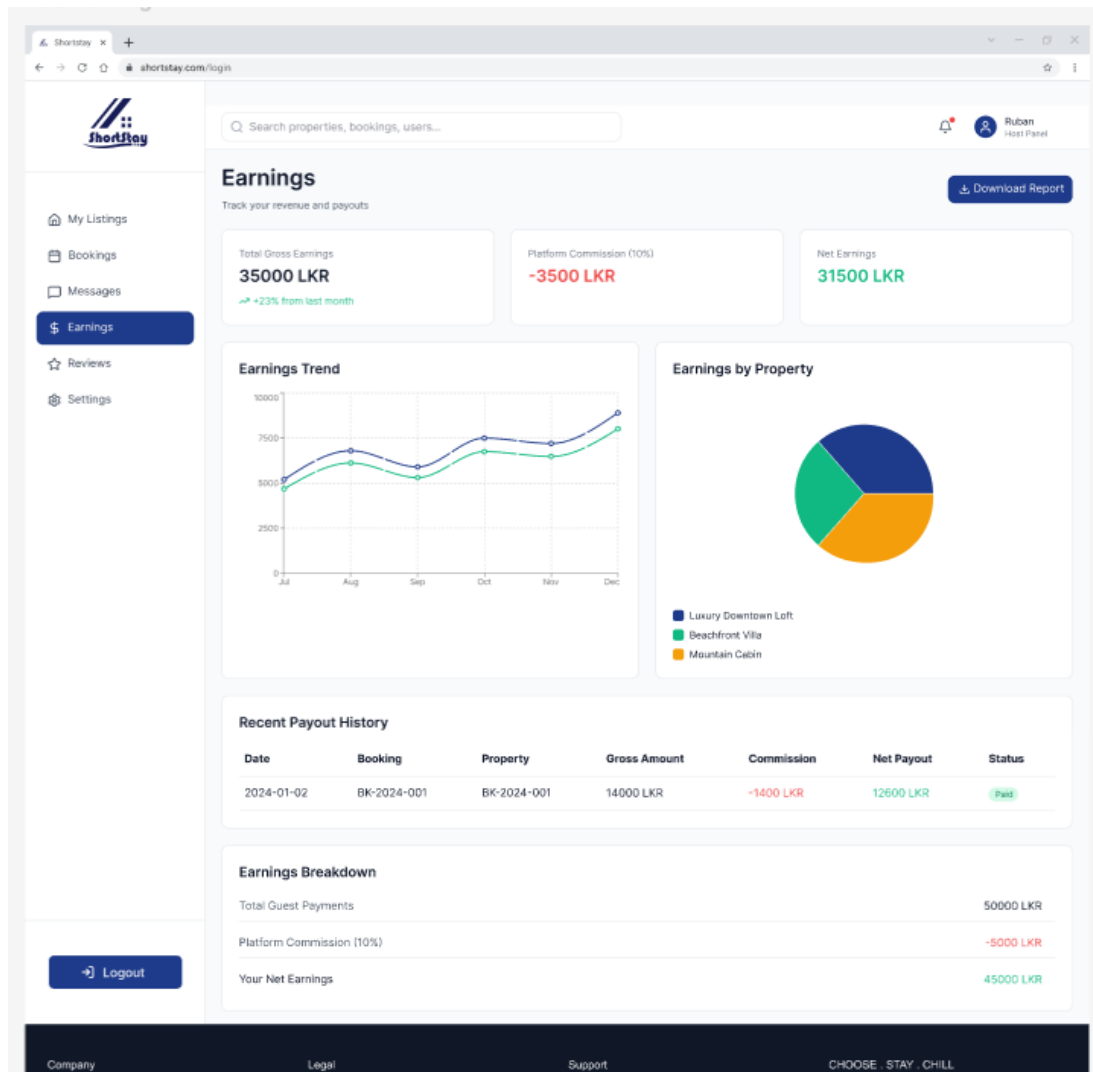


Figure 9 - Host Dashboard

4.4 Chapter Summary

The following details have been outlined in this chapter with regard to the design solution for the ShortStay application:

- Role-based dashboards for Guest, Host, and Admin
- UI/UX design created using the Figma tool with a focus on usability and responsiveness

The frontend design is visually consistent with the Figma design, providing a robust base for the backend development process.

5 Conclusion and Future Work

5.1 Introduction

This chapter presents the overall progress of the ShortStay project up to the current stage. It summarizes the achievements made during the requirement gathering and system design phases. It also discusses the lessons learned, limitations faced by the team, and the future work required to successfully complete the project.

5.2 Achievements to Date

Key achievements include:

1. Requirement Gathering and Analysis
 - Identified all five user roles: Guest, Host, Admin, Payment Manager, and Field Inspector
 - Completed requirement gathering and system analysis
 - Completed Use Case Diagram including main and sub functionalities
 - Designed the Entity Relationship (ER) Diagram for the database
 - Developed UML Class Diagram
 - Selected appropriate tools and technologies for development
2. UI/UX Design
 - Designed UI/UX layouts for all user roles
 - Developed user-friendly interface structure for system navigation
 - Improved interface designs based on team discussions and feedback
3. Documentation progress monitoring
 - Prepared Project Proposal and two progress reports
 - Created Agile-based Gantt Chart
 - Assigned roles and responsibilities among team members
 - Conducted regular meetings to monitor progress

5.3 Lessons Learned

- Importance of proper requirement analysis before development
- Effective communication improves team coordination
- Agile methodology helps manage tasks efficiently
- System diagrams help to understand the system structure clearly
- Time management is critical due to academic workload
- Understanding complex workflows, such as payment and verification, requires detailed planning
- Team collaboration is essential for successful project completion

5.4 Limitations

- Limited experience in some technologies during the early stages
- Difficulty in understanding complete system workflows initially
- Time constraints due to academic workload and examinations
- Challenges in coordinating physical meetings with all team members

5.5 Future Work

1. Complete frontend and backend development
2. Perform system testing (unit testing, integration testing, and user acceptance testing)
3. Identify and fix bugs in both frontend and backend systems
4. Improve system performance and optimize database operations
5. Enhance UI/UX based on user feedback
6. Implement additional security improvements
7. Deploy the system on a suitable hosting platform
8. Prepare final documentation and project presentation

5.6 Conclusion

In conclusion, the ShortStay project has successfully completed requirement analysis and system design. The team has designed all interfaces, achieving the main functionalities of the system.

Although some challenges were encountered, such as time constraints and technical limitations, the team was able to overcome them through proper planning and teamwork. The project is now ready for development, testing, and deployment phases.

5.7 Chapter Summary

This chapter provided an overview of the project progress, including completed work, lessons learned, and limitations. It also described the future work required to finalize the system.

Overall, the ShortStay project is progressing successfully and is moving towards completion.

- The End -